

Statistics 250, Section 003, Quiz 3 (150 Points)

April 14, 1999, Dr. Jürgen Symanzik

Your Name: _____

Question 1: Samples and Populations (55 Points)

Below are the ages of all 40 blood donors from a computer company in Fairfax, Virginia, that visited a bloodmobile unit at their company on a particular day. We consider these 40 blood donors as our population of interest for the remainder of this question. And here is the data:

35	53	61	31	21	42	23	29	35	37
39	58	27	64	27	31	36	48	41	22
37	35	42	32	43	34	59	50	38	43
31	30	41	37	29	45	23	56	46	41

1. Obviously, **35%** of the blood donors who visit the bloodmobile at this company this particular day fall into the 30 to 39 years age group. Is this number (i.e., **35%**) a parameter or a statistic? (**5 Points**)
2. Based on this data, do you think it is correct to state that about 35% of all blood donors anywhere in the United States fall into the 30 to 39 years age group? Explain, using the appropriate statistical vocabulary. (**10 Points**)

3. A local newspaper reporter from Broadside wants to learn more about some of these blood donors and their motivation to come to the bloodmobile. Since the reporter once took a Stat 250 class, she thinks that it might be best to randomly select people for the interview rather than taking the first people that show up for the interview. The reporter wants to talk to 5 people and the bloodmobile operator hands her a list of those 40 people that participated.

Explain how you would proceed if you were the reporter, using the appropriate statistical vocabulary. A random digits table has been attached. Which persons (indicate the numbers or circle the corresponding age on the previous page) do you finally select for your interview? **(20 Points)**

4. Based on your 5 persons from (3.) above, calculate the sample proportion $\hat{p}$ of blood donors at this company that falls into the 30 to 39 years age group. Is this number a parameter or a statistic? **(5 Points)**

5. Fill in the missing 2 words:

$\hat{p}$ is a _____ that can be used as an estimate for the (unknown) population proportion p . **(5 Points)**

6. Unfortunately, our reporter does not have the time to calculate the mean age for those 5 persons from (3.) above. Instead, she asks her assistant (a student who dropped out of Stat 250 after the first week) to calculate the mean age of these 5 persons. The student comes up with 22 years. Should the reporter be suspicious about this number or is this just a possible effect of sampling? Explain, using the appropriate statistical vocabulary. Note that the reporter may be using different persons than the 5 persons you used in the previous parts of this question. **(10 Points)**

Question 2: Residual Plots (24 Points)

Look at these 4 residual plots from G.A.F. Seber's "Linear Regression Analysis" book and answer the following two questions:

1. Which of these residual plot(s) represent(s) satisfactory pattern(s) that show that your selected statistical model (e.g., a least squares regression line) describes the data reasonably well.
2. For each of the residual plots that does not represent a satisfactory pattern, indicate the type of problem that occurs with the data in that particular case.

Question 3: Probability Distributions (**31 Points**)

For homework assignment #8, Amanda reported the following point distribution for those 50 Stat 250 students that turned in this particular homework assignment:

# students	points
1	9
1	10
0	11
1	12
0	13
2	14
2	15
1	16
5	17
2	18
10	19
25	20

We introduce a random variable Y that represents the points for homework assignment #8.

1. Indicate the probability distribution $p(y)$ for Y in tabular form. (**10 Points**)

2. Add the values for $F(y)$, i.e., the cumulative probability function for Y , to your table in (1.) above. What is $P(Y \leq 18)$? (**11 Points**)

3. Calculate $\mu = E(Y)$, i.e., the mean (expected value) of Y . (**10 Points**)

Question 4: Residuals (40 Points)

Students from Stat 250 wanted to determine the thickness of paper sheets used for the photocopying machines in S&T II. They grabbed a random number of sheets, counted the number of sheets, and measured the thickness of this pile of paper. Here is the data:

$x = \# \text{ Sheets}$	$y = \text{Thickness [mm]}$
149	9.2
117	7.2
165	10.0
145	8.8
155	9.5
94	6.2
413	24.8
111	7.3
238	14.1
84	4.8

We found that a least squares regression line of the form

$$\hat{y} = 0.3290 + 0.0590x$$

seems to fit the data reasonably well.

1. Calculate the residuals based on the least squares regression line and data given above. **(10 Points)**

2. Draw at least two different residual plots (make clear what you plot on the x-axis each time). One of your two residual plots should reflect the fact that two different measuring techniques have been used — the first 5 measurements have been made according to technique 1, the last 5 measurements have been made according to technique 2. State your conclusions. **(20 Points)**

3. We have seen in (2.) above that the measuring technique might influence the outcome of this study. List at least two other lurking variables that could have an influence on the outcome of the measurements of the photocopying paper but do not show up in this data set. **(10 Points)**